



COMPETENCY STANDARD

FOR

SHIP PIPING

(SHIPBUILDING SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for Ship Piping serve as a foundational document for the development of curricula, teaching materials, and assessment tools specifically tailored to the shipbuilding sector. This document ensures that training activities align with industry requirements, enabling individuals who successfully meet the established standards through assessment to become qualified professionals in relevant roles within the sector.

Developed under the Skills for Industry Competitiveness and Innovation Program (SICIP), this document addresses the skills requirements critical to Ship Piping. It is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION:

The **Skills for Industry Competitiveness and Innovation Program (SICIP)** has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from Construction industry in consultative workshop.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:**Generic Competencies (50 Hrs.)**

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SBD-SHP-01-G	Apply Occupational Health and Safety (OHS) Practice in the Workplace	1. Identify OHS policies and procedures. 2. Apply personal health and safety practices. 3. Report hazards and risks 4. Respond to emergencies.	20
SICIP-SBD-SHP-02-G	Carry Out Workplace Interaction	1 Interpret workplace communication and etiquette. 2 Read and understand workplace documents. 3 Participate in workplace meetings and discussions. 4 Practice professional ethics at work	20
SICIP-SBD-SHP-03-G	Operate In a Team Environment	1 Identify team goals and work processes. 2 Identify own role and responsibilities within team. 3 Communicate and co-operate with team members. 4 Practice problem solving within the team.	10
Total Hour			50

Sector Specific Competencies (Total 40 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SBD-SHP-01-S	Apply Green Practices in Ship Building Sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	15
SICIP-SBD-SHP-02-S	Use Hand Tools and Power Tools	1. Identify and inspect hand and power tools 2. Use hand tools properly and safely 3. Operate power tools properly and safely. 4. Clean and maintain hand and power tools	25
Total Hour			40

Occupation Specific Competencies (Total 270 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-SBD-SHP-01-O	Identify Basic Ship Piping Work	1. Identify key tasks of a ship pipe fitter 2. Interpret piping system 3. Understand drawings, symbols and specifications 4. Identify color codes and standards	20
SICIP-SBD-SHP-02-O	Recognize Pipe and Pipe Fittings Components	1. Identify the different types of pipes 2. Identify different types of valves 3. Identify different types of fittings 4. Identify different types of pumps 5. Identify different types of hangers/support and fasteners	20
SICIP-SBD-SHP-03-O	Perform Welding Works	1. Identify types of welding and welding terminology 2. Identify different types of welding technique 3. Carry out arc welding 4. Carry out tack welding	40
SICIP-SBD-SHP-04-O	Perform Pipe Fabrication Works	1. Cut, thread and bevel pipes 2. Perform pipe bending with different angle 3. Clean and test piping 4. Secure supports, hangers and guides	120
SICIP-SBD-SHP-05-O	Perform Installation of Piping System	1. Install fresh water piping system 2. Install sea water piping system 3. Install fuel piping system 4. Install hydraulic piping system	70
Total Hours			270 Hrs.

COMPETENCY STANDARD: SHIP PIPING

Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICE IN THE WORKPLACE	Nominal Duration: 20 hrs.	Unit Code: SICIP-SBD-SHP-01-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, applying personal health and safety practices, reporting hazards and risks and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 <u>OHS policies</u> and safe operating procedures are interpreted 1.2 Safety signs and symbols are identified and followed. 1.3 Response, evacuation procedures and other contingency measures are interpreted correctly.
2. Apply personal health and safety practices	2.1 OHS policies and procedures are interpreted in the workplace including <u>personal protective equipment (PPE)</u> . 2.2 Common health issues are recognized. 2.3 Common safety issues are identified and applied.
3. Report hazards and risks	3.1 Hazards and risks are identified. 3.2 Hazards and risks assessment and controls are interpreted 3.3 Hazards and risk assessment are reported
4. Respond to emergencies	4.1 Respond to alarms and warning devices. 4.2 <u>Emergency response plans and procedures</u> are responded to. 4.3 <u>First aid procedures</u> during emergency situations are identified.

Range of variables:

Variables	Range (may include but not limited to)
1. OHS policies	1.1 Organizational OHS policies 1.2 International OHS requirements 1.3 Fire safety rules and regulations
2. Personal protective	2.1 Safety glasses

equipment	2.2	Ear plugs
	2.3	Gloves
	2.4	Apron
	2.5	Helmet
	2.6	Mask
	2.7	Safety shoes
3. Emergency response plans and procedures	3.1	Firefighting procedures
	3.2	Earthquake response procedures
	3.3	Emergency response plans and procedures
	3.4	Medical and first aid
4. First aid procedure	4.1	Washing of open wound
	4.2	Washing chemically infected area
	4.3	Applying bandage
	4.4	Taking appropriate medicine

Curricular Content Guide

1. Underpinning knowledge	1.1	Workplace OHS policies and procedures
	1.2	Work safety procedures
	1.3	Emergency response procedures:
	1.3.1	Firefighting
	1.3.2	Earthquake response
	1.3.3	Accident response
	1.4	Types of hazards (biological, chemical and physical) and their effects
	1.5	OHS awareness
	1.6	Personal protective equipment (PPE)
	1.7	Malfunctions and resolutions
2. Underpinning skill	2.1	Identifying OHS policies and procedures
	2.2	Applying personal health and safety practices
	2.3	Reporting hazards and risks
	2.4	Responding to emergencies
3. Underpinning Attitudes	3.1	Commitment to occupational health and safety
	3.2	Promptness in carrying out activities
	3.3	Sincere and honest to duties
	3.4	Environmental concerns
	3.5	Eagerness to learn
	3.6	Tidiness and timeliness
	3.7	Respect for rights of peers and seniors in the workplace
	3.8	Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided:	
	4.1	Workplace (simulated or actual)
	4.2	Workplace documents, signs and symbols
	4.3	Codes of conduct
	4.4	Projector
	4.5	Learning manual

	4.6 Tools, equipment and facilities appropriate to the process or activities.
	4.7 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Identified OHS policies and procedures 1.2 Applied personal health and safety practices (including PPE) 1.3 Reported hazards and risks 1.4 Responded to emergencies.
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: CARRY OUT WORKPLACE INTERACTION	Nominal Duration: 20 hrs.	Unit Code: SICIP-SBD-SHP-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to carry out workplace interaction. It specifically includes the tasks of interpreting workplace communication and etiquette, reading and understanding workplace documents, participating in workplace meetings and discussions and practicing professional ethics at work.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret workplace communication and etiquette	1.1 Workplace codes of conduct are interpreted as per organizational guidelines. 1.2 Appropriate lines of communication are maintained with supervisors and colleagues. 1.3 Workplace interactions are conducted in a <u>courteous manner</u> to gather and convey information. 1.4 <u>Workplace procedures and matters</u> are comprehended.
2. Read and understand workplace documents	2.1 Workplace documents are interpreted correctly. 2.2 Visual information/symbols/signage are understood correctly and followed. 2.3 Specific and relevant information are accessed from <u>appropriate sources</u> . 2.4 Appropriate medium is used to transfer information and ideas.
3. Participate in workplace meetings and discussions	3.1 Team meetings are attended on time. 3.2 Meeting procedures and etiquette are followed. 3.3 Active participation is ensured, opinions are expressed and heard. 3.4 Inputs are provided and interpreted in line with the meeting purpose.
4. Practice professional ethics at work	4.1 Responsibilities as a team member are performed. 4.2 Tasks are performed in accordance with workplace procedures. 4.3 Confidentiality is maintained. 4.4 Inappropriate and conflicting situations are avoided.

Range of variables:

Variables	Range (may include but not limited to)
1. Courteous manner	1.1 Effective questioning 1.2 Active listening

	1.3 Speaking skills 1.4 Writing skill 1.5 Email etiquette
2. Workplace procedures and matters	2.1 Notes 2.2 Arranging a meeting 2.3 Agenda 2.4 Simple reports such as progress and incident reports 2.5 Job sheets 2.6 Operational manuals 2.7 Brochures and promotional material 2.8 Visual and graphic materials 2.9 Standards 2.10 OHS information 2.11 Signs
3. Appropriate sources	3.1 Human Resources (HR) Department 3.2 Managers 3.3 Supervisors 3.4 Management Information System (MIS)

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace communication and etiquette 1.2 Workplace documents, signs and symbols 1.3 Meeting procedure and etiquette 1.4 Professional ethics
2. Underpinning skill	2.1 Demonstrating workplace communication and etiquette 2.2 Interpreting workplace instructions and symbols 2.3 Demonstrating active participation in workplace meeting 2.4 Applying professional ethics at work
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Learning manual 4.5 Tools, equipment and facilities appropriate to the process or activities. 4.6 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Interpreted workplace communication and etiquette 1.2 Interpreted workplace instructions and symbols 1.3 Performed active participation in workplace meetings
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: OPERATE IN A TEAM ENVIRONMENT	Nominal Duration: 10 hrs.	Unit Code: SICIP-SBD-SHP-03-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to operate in a team environment. It specifically includes the tasks of identifying team goals and work processes, identifying own role and responsibilities within team, communicating and cooperating with team member, and practicing problem solving within the team.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes	1.1. Roles and objectives of the team are identified and interpreted. 1.2. Roles and responsibilities of team members and work processes are identified and interpreted.
2. Identify own role and responsibilities within team	2.1. Personal role and responsibilities are identified within the team environment. 2.2. Importance of relationships within and outside the team are interpreted .
3. Communicate and co-operate with team members	3.1. Other teammates' tasks are identified and support provided when requested. 3.2. The team is encouraged through <u>sharing information</u> or expertise, working together to solve problems, and putting team success first. 3.3. Views and opinions of other team members are interpreted and respected.
4. Practice problem solving within the team	4.1. Problems faced at the individual and team level are identified and showed insight into the root-causes of the problems. 4.2. A range of solutions and courses of action are identified together with benefits, costs, and risks associated with each. 4.3. The good ideas of others to help develop solutions are recognized and advice sought from those who have solved similar problems. 4.4. It is looked beyond the obvious and not stopped at the first answers.

Range of variables:

Variables	Range (may include but not limited to)
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1. Sharing information	1.1. Agenda 1.2. Minutes 1.3. Progress and incident reports 1.4. Operational manuals 1.5. Visual and graphic materials 1.6. Emails and SMS 1.7. Policy, procedure and standards 1.8. OHS information
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Curricular Content Guide

1. Underpinning knowledge	1.1. Team goals and work processes 1.2. Roles and responsibilities of the team 1.3. Finding problems and solving them 1.4. Importance of views and opinions of other team members
2. Underpinning skill	2.1. Identifying own role and responsibilities within team 2.2. Communicating and co-operating with team members 2.3. Demonstrating problem solving within the team
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Eagerness to learn 3.5. Tidiness and timeliness 3.6. Environmental concerns 3.7. Respect for rights of peers and seniors at workplace 3.8. Communication with peers and seniors at workplace
4. Resource implications	The following resources must be provided: 4.1. Workplace (simulated or actual) 4.2. Workplace documents, Projector 4.3. Learning manual 4.4. Tools, equipment and facilities appropriate to the process or activities. 4.5. Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Identified own role and responsibilities within team 1.2 Communicated and co-operated with team members 1.3 Demonstrated problem solving within the team
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning

	2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: APPLY GREEN PRACTICES IN SHIPBUILDING SECTOR	Nominal Duration: 15 hrs.	Unit Code: SICIP-SBD-SHP-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply green practices in shipbuilding sector. It specifically includes the tasks of interpreting green concepts, minimizing resources in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 <u>Green concepts in the shipbuilding sector</u> are understood. 1.2 <u>Energy-efficient technologies</u> are recognized. 1.3 <u>Use of eco-friendly materials</u> is understood. 1.4 <u>Pollution control measures</u> are interpreted. 1.5 <u>Regulatory compliance</u> with regulations and conventions is identified. 1.6 <u>Energy recovery systems</u> are identified as methods to improve energy efficiency.
2. Minimize resources use in the workplace	2.1 <u>Resource usage in shipbuilding processes</u> is identified. 2.2 Water consumption methods are identified through the installation of efficient water management systems. 2.3 <u>Process improvements</u> are understood. 2.4 Compliance with environmental regulations and industry standards is interpreted.
3. Implement waste management practices	3.1 <u>The types of waste in shipbuilding</u> are identified and described. 3.2 <u>Environmental impacts of improper waste disposal</u> in shipbuilding are understood. 3.3 <u>Principles of waste hierarchy</u> are understood for sustainable waste management. 3.4 <u>Importance of safe storage and containment of waste materials</u> is understood.

Range of Variables

Variable	Range
	May include but not limited to:
1. Green concepts in the shipbuilding sector	1.1 Eco-friendly Hull Design 1.2 Use of Sustainable and Recyclable Materials 1.3 Waste Reduction and Management Practices

	1.4 Emission Control and Air Pollution Reduction 1.5 Ballast Water Treatment and Marine Ecosystem Protection
2. Energy-efficient technologies	2.1 Hybrid and Electric Propulsion Systems 2.2 Advanced Hull Coatings to Reduce Drag 2.3 Waste Heat Recovery Systems 2.4 Air Lubrication Systems 2.5 Energy-efficient LED Lighting and Automation 2.6 Smart Energy Management and Monitoring Systems
3. Use of eco-friendly materials	3.1 Recycled Steel and Aluminum 3.2 Biodegradable Composites and Polymers 3.3 Sustainably Sourced Timber 3.4 Non-toxic Anti-fouling Coatings 3.5 Natural Fiber Reinforced Plastics 3.6 Low VOC (Volatile Organic Compounds) Paints and Sealants
4. Pollution control measures	4.1 Prevention of water 4.2 Prevention of air 4.3 Prevention of noise pollution
5. Regulatory compliance	5.1 International Maritime Organization (IMO) Regulations 5.2 Environmental Protection Agency (EPA) Standards 5.3 Marine Pollution (MARPOL) Convention Compliance 5.4 Occupational Health and Safety (OHS) Regulations 5.5 Ship Recycling and Basel Convention Guidelines 5.6 Local and National Environmental Permitting
6. Energy recovery systems	6.1 Waste Heat Recovery Boilers (WHRB) 6.2 Thermoelectric Energy Conversion Systems 6.3 Exhaust Gas Recirculation (EGR) Systems 6.4 Regenerative Braking for Hybrid Propulsion
7. Resource usage in shipbuilding processes	7.1 Raw Material Consumption (Steel, Aluminum, Composites) 7.2 Water Usage in Fabrication and Painting 7.3 Energy Consumption During Manufacturing 7.4 Use of Recycled and Sustainable Materials
8. Process improvements	8.1 Automation and Robotics in Assembly 8.2 Modular Construction and Prefabrication 8.3 Digital Twin and 3D Modeling Integration 8.4 Just-In-Time (JIT) Inventory Management 8.5 Enhanced Quality Control and Testing Procedures
9. Types of waste in shipbuilding	9.1 Hazardous wastes 9.1.1 Paints 9.1.2 Solvents 9.1.3 Oils 9.1.4 Asbestos 9.2 Non-hazardous wastes 9.2.1 Metal scraps 9.2.2 Wood and general refuse

10. Environmental impacts of improper waste disposal	10.1 Including soil contamination 10.2 Water pollution, and air quality degradation, are understood.
11. Principles of waste hierarchy	11.1 Reduction 11.2 Reuse 11.3 Recycling 11.4 Recovery 11.5 Disposal
12. Importance of safe storage and containment of waste materials	12.1 Prevent leaks 12.2 Prevent spills 12.3 Environmental contamination

Curricular Content Guide

1. Underpinning Knowledge	1.1 Fundamental Green Concepts and Sustainability Principles in Shipbuilding 1.2 Overview of Energy-Efficient Technologies and Their Applications 1.3 Types and Benefits of Eco-Friendly Materials in Shipbuilding 1.4 Key Pollution Control Measures and Their Implementation 1.5 Regulatory Frameworks, Conventions, and Compliance Requirements 1.6 Energy Recovery Systems and Their Role in Enhancing Efficiency 1.7 Resource and Water Usage Management in Shipbuilding Processes 1.8 Understand of environmental Impacts, and Sustainable Practices
2. Underpinning Skills	2.1 Applying Green Concepts in Shipbuilding Practices 2.2 Evaluating and Selecting Energy-Efficient Technologies 2.3 Identifying and Using Eco-Friendly Materials Effectively 2.4 Implementing Pollution Control Measures Onsite 2.5 Interpreting and Ensuring Regulatory Compliance 2.6 Operating and Maintaining Energy Recovery Systems 2.7 Monitoring and Managing Resource and Water Usage 2.8 Improving Shipbuilding Processes for Environmental Sustainability 2.9 Classifying, Handling, and Safely Storing Shipbuilding Waste
3 Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns

	3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted green concepts 1.2 minimizing resources use in the workplace 1.3 implementing waste management practices
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: USE HAND AND POWER TOOLS	Nominal Duration: 25 hrs.	Unit Code: SICIP-SBD-SHP-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to use hand and power tools in the workplace. It specifically includes the task of identifying and inspecting hand and power tools, using hand tools properly and safely, operating power tools properly and safely and cleaning and maintaining hand and power tools.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify and inspect hand and power tools	1.1 Appropriate hand and power tools are identified. 1.2 Application of hand and power tools is recognized. 1.3 Usability of hand and power tools is checked and verified.
2. Use hand tools properly and safely	2.1 Appropriate <u>hand tools</u> are selected. 2.2 Safety precautions are ensured before using hand tools. 2.3 Unsafe or faulty hand tools are identified and marked for repair. 2.4 <u>Measuring tools</u> are checked and calibrated before use. 2.5 Use hand tools properly and safely to perform work activity.
3. Operate power tools properly and safely	3.1. Appropriate <u>power tools</u> are selected. 3.2. Power supply outlet and electrical cord are inspected and confirmed safe for use in accordance with established workplace safety requirements. 3.3. Safety precautions are ensured before using power tools in accordance with manufacturer's operating specification. 3.4. Proper sequence of operation applied for using power tools. 3.5. Unsafe or faulty power tools are identified and marked for repair. 3.6. Operate power tools properly and safely to perform work activity.
4. Clean and maintain hand and power tools	4.1. Dust and foreign matter is removed from hand and power tools in accordance to workplace standards. 4.2. Condition of hand and power tools is checked after use and reported. 4.3. Appropriate lubricant is applied after use and prior to storage.

	4.5. Defective hand and power tools are inspected and repaired or replaced. 4.6. Hand and power tools are stored and secured in accordance with workplace requirements.
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Range of Variables

Variable	Range May include but not limited to:
1. Hand tools	1.1 Hammer 1.2 Bench vice 1.3 Files 1.4 Wrenches 1.5 Pliers 1.6 Scriber 1.7 Scraper Screw 1.8 Surface plate 1.9 Gauge 1.10 Tap sets 1.11 Die sets 1.12 Hacksaw 1.13 Spanners 1.14 Vice grip 1.15 Wire cutters 1.16 Clamps 1.17 Jacks
2. Power tools	2.1 Drills 2.2 Rivet gun 2.3 Grinders 2.4 Pneumatic wrenches 2.5 Press machine 2.6 Cutting 2.7 Saws 2.8 Soldering iron
3. Measuring tools	3.1 Micrometer 3.2 Measuring tape 3.3 Hose level 3.4 Water level 3.5 Calipers 3.6 Steel rule 3.7 Meter rule 3.8 Spirit level 3.9 Protractor 3.10 Tri-square 3.11 Gauges

Curricular Content Guide

1. Underpinning Knowledge	1.1 Information on hand and power tools, their functions and use 1.2 Procedures for safely using hand and power tools
2. Underpinning Skills	2.1 Identifying hand, power and measuring tools 2.2 Following safety precautions when using hand, power and measuring tools 2.3 Using hand and measuring tools correctly and safely in accordance with manufacturer's operating specification 2.4 Operating power tools correctly and safely in accordance with manufacturer's operating specification 2.5 Cleaning and maintaining hand and power tools after use 2.6 Applying appropriate lubricant on hand and power tools after use and prior to storing
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process or activities. 4.8 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified and selected appropriate hand and power tools for work to be performed 1.2 identified and used measuring and testing tools appropriate to work activity 1.3 followed safety precautions when using hand and power tools 1.4 operated power tools safely and pursuant to manufacturer's operating specification
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	1.5 performed cleaning and maintenance of hand and power tools after use and prior to storing
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: IDENTIFY BASIC SHIP PIPING WORK	Nominal Duration: 20 Hrs.	Unit Code: SICIP-SBD-SHP-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to identify basic ship piping work. It specifically includes the task of identifying key tasks of pipe fitter, interpreting ship piping systems, understanding drawings, symbols and specifications, and identifying color codes and standards.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify key tasks of a ship pipe fitter	1.1. Work instructions and safety protocols related to ship piping operations are interpreted. 1.2. Key tasks of a ship pipe fitter are identified based on shipbuilding and maintenance requirements. 1.3. <u>Roles and responsibilities of a pipe fitter</u> are explained in the context of marine piping systems. 1.4. Pipe fitting tasks are listed and described.
2. Interpret piping system	2.1 <u>Ship piping systems</u> are identified and explained according to function and layout. 2.2 <u>Major components of the piping system</u> are identified and described. 2.3 Functions of various piping systems are interpreted based on ship operations.
3. Understand drawings, symbols and specifications	3.1 Types of <u>piping drawings</u> are explained as per job requirement. 3.2 Components of spool sheet drawing are identified. 3.3 <u>Pipe fitting symbols</u> are identified and defined. 3.4 Information contained in <u>specifications of piping work</u> is explained.
4. Identify color codes and standards	4.1 <u>Color codes</u> used in ship piping systems are identified and described based on job specifications. 4.2 Standard color markings for various systems are recognized as per industry regulations. 4.3 Color coding charts and guidelines are interpreted according to shipyard or international standards.

Range of Variables

Variable	Range (Includes but not limited to):
1. Roles and responsibilities of a pipe fitter	1.1 Read and interpret shipbuilding blueprints and pipe system drawings 1.2 Cut, bend, and assemble pipes to required

	specifications 1.3 Install and secure pipes for steam, water, fuel, and hydraulic systems 1.4 Fit and weld pipe components using various methods. 1.5 Conduct pressure tests to check for leaks or weak joints 1.6 Ensure proper alignment and support of piping systems 1.7 Follow safety procedures and shipyard regulations 1.8 Maintain tools and equipment in good condition 1.9 Keep work area clean and organized
2. Ship piping system	2.1 Ballast System 2.2 Bilge System 2.3 Fuel Oil System 2.4 Lubricating Oil System 2.5 Fresh Water System 2.6 Sea Water Cooling System 2.7 Fire Fighting System 2.8 Compressed Air System 2.9 Steam System 2.10 Hydraulic System 2.11 Sewage System 2.12 Cargo Piping System 2.13 Ventilation and Air Conditioning (HVAC) System 2.14 Exhaust System
3. Major Components of the piping system	3.1 Pipes 3.2 Valves 3.3 Flanges 3.4 Fittings 3.5 Pumps 3.6 Strainers/Filters 3.7 Expansion Joints 3.8 Gaskets 3.9 Supports and Hangers 3.10 Instrumentation 3.11 Tanks and Reservoirs 3.12 Insulation
4. Piping drawings	4.1 P&ID (Piping & Instrumentation Diagram) 4.2 Isometric Drawing 4.3 General Arrangement (GA) Drawing 4.4 Detailed Piping Plan 4.5 Fabrication/Spool Drawing
5. Pipe fitting symbols	5.1 Elbow (90° or 45°) 5.2 Tee (T-joint) 5.3 Reducer (Concentric/Eccentric) 5.4 Coupling 5.5 Union 5.6 Flange 5.7 Different type of valves

	5.8 Pump 5.9 Strainer 5.10 Expansion Joint
6. Specifications of piping work	6.1 Pipe Material 6.2 Pipe Size & Schedule 6.3 Pressure Rating 6.4 Temperature Rating 6.5 Welding Standards 6.6 Corrosion Protection 6.7 Testing Requirements 6.8 Insulation Requirements 6.9 Support and Hanger Types 6.10 Flange and Fitting Standards
7. Color Codes	7.1 Sea Water – Green 7.2 Fresh Water – Light Blue 7.3 Fuel Oil – Brown 7.4 Lubricating Oil – Yellow with Brown Band 7.5 Hydraulic Oil – Yellow 7.6 Steam – Silver/Grey 7.7 Air Line – White 7.8 Fire Fighting – Red 7.9 Bilge Water – Black 7.10 Ballast Water – Green with Black Band 7.11 Sewage – Black with Brown Band 7.12 Cargo (Liquid/Gas) – Orange

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify key tasks of a ship pipe fitter based on shipbuilding and maintenance requirements. 1.2 Explain the roles and responsibilities of a pipe fitter in marine piping systems. 1.3 Identify and explain ship piping systems according to their function and layout. 1.4 Describe major components of the piping system. 1.5 Interpret the functions of various piping systems based on ship operations. 1.6 Identify components of spool sheet drawings. 1.7 Identify and define pipe fitting symbols. 1.8 Explain the information contained in specifications of piping work. 1.9 Identify and describe color codes used in ship piping systems based on job specifications. 1.10 Interpret color coding charts and guidelines according to shipyard or international standards.
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2. Underpinning Skills	2.1 Demonstrating identification of key tasks of a ship pipe fitter based on shipbuilding and maintenance needs. 2.2 Applying knowledge to explain roles and responsibilities of a pipe fitter in marine piping systems. 2.3 Fixing pipe fitting tasks by listing and describing the required activities. 2.4 Maintaining understanding of ship piping systems and explaining them according to function and layout. 2.5 Servicing the identification and description of major components of the piping system. 2.6 Performing interpretation of piping system functions based on ship operations. 2.7 Demonstrating explanation of piping drawings types based on job requirements. 2.8 Applying the identification and definition of pipe fitting symbols.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety. 3.2 Promptness in carrying out activities. 3.3 Sincere & honest to duties. 3.4 Eagerness to learn. 3.5 Tidiness & timeliness. 3.6 Environmental concerns. 3.7 Respect for rights of peers and seniors at workplace. 3.8 Communication with peers and seniors at workplace
4 Resource Implications	The following resources must be provided: 4.1 workplace (actual or simulate) 4.2 materials relevant to proposed activities 4.3 tools and equipment 4.4 piping drawings

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified key tasks of a ship pipe fitter 1.2 interpreted piping system 1.3 understood drawings, symbols and specifications 1.4 identified color codes and standards
2. Methods of Assessment	2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: RECOGNIZE PIPE AND PIPE FITTING COMPONENTS	Nominal Duration: 20 Hrs.	Unit Code: SICIP-SBD-SHP-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to identify pipe and pipe fitting components. It specifically includes the task of identifying the different types of pipes, identifying different types of valves, identifying different types of fittings, identifying different types of pumps and identifying different types of hangers/support and fasteners.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify different types of pipes	1.1 <u>Different types of pipes</u> are identified based on material and usage. 1.2 Steel pipe dimensions chart is identified based on system requirements. 1.3 <u>Material characteristics of GI pipe</u> are described. 1.4 <u>Material characteristics of MS pipe</u> are described. 1.5 Pipe standards and markings are interpreted in accordance with industry codes.
2. Identify different types of valves	2.1 <u>Different types of valves</u> are identified based on function and design. 2.2 Valve functions are described according to technical specifications. 2.3 Installation methods of valves are explained following standard operating procedures.
3. Identify different types of fittings	3.1 <u>Different types of pipe fittings</u> are identified based on shape and function. 3.2 Connection methods for fittings are explained according to system requirements. 3.3 Fitting standards and specifications are interpreted based on engineering drawings or manuals. 3.4 Incompatible or defective fittings are identified.
4. Identify different types of pumps	4.1 <u>Different types of pumps</u> are identified based on their design and application. 4.2 Main components of pumps are identified and their functions are described. 4.3 Pump specifications and operational symbols are interpreted according to technical documents. 4.4 Faulty or incompatible pumps are identified.
5. Identify different types of hangers/support and fasteners	5.1 <u>Different types of hangers, supports, and fasteners</u> are identified based on design and use. 5.2 Pipe supporting and fastening methods are explained.

	5.3 Standards and load capacities of supports and fasteners are interpreted from technical guidelines.
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Range of Variables

Variable	Range (Includes but not limited to):
1. Different types of pipes	1.1 Galvanized iron (G.I) 1.2 Mild steel (M.S) 1.3 Stainless steel (S.S) 1.4 Copper 1.5 Alloy steel 1.6 Carbon steel 1.7 Aluminium
2. Material characteristics of GI pipe	2.1 Made from carbon steel coated with zinc. 2.2 Zinc coating prevents rust and corrosion. 2.3 Long-lasting in moisture-rich environments. 2.4 High tensile strength due to steel core. 2.5 Heavier than non-galvanized pipes. 2.6 Can handle moderate temperatures (up to 200°C/392°F). 2.7 Easy to thread, weld, and join. 2.8 Affordable compared to stainless steel. 2.9 Safe for water distribution if coating is intact.
3. Material characteristics of MS pipe	3.1 Made from mild steel, a low-carbon (0.05% to 0.25%) steel 3.2 High strength and durability due to the steel composition. 3.3 Good weldability and malleability, making it easy to fabricate. 3.4 Corrosion-prone if not coated or protected 3.5 Lightweight compared to other steel pipes. 3.6 Cost-effective and widely used in fuel and engine oil line, construction, plumbing, and structural applications. 3.7 Easily machinable and can be shaped into different forms. 3.8 Moderate temperature resistance 3.9 Not suitable for high-pressure systems without reinforcement. 3.10 Good for low to medium pressure applications.
4. Different types of valves	4.1 Globe 4.2 Gate 4.3 Screw down non-return 4.4 Butterfly 4.5 Relief/safety 4.6 Ball 4.7 Swing check 4.8 Remote control 4.9 Self-closing

	4.10 Air head
5. Different types fittings	5.1 Elbow 5.1.1 Long bend 5.1.2 Short bend 5.2 Tee 5.2.1 Threaded Tee 5.2.2 Y and Cross Tee 5.3 Flange with nut bolt 5.4 Reducer 5.5 Socket 5.6 Union 5.7 Nipple 5.8 Socket and bon plug
6. Different types of pumps	6.1 Centrifugal Pump 6.2 Gear Pump 6.3 Screw Pump 6.4 Reciprocating Pump 6.5 Submersible Pump
7. Different types of hangers/support and fasteners	Hangers/support: 7.1 Clevis hangers 7.2 U-bolt clamps 7.3 Pipe straps 7.4 Riser clamps 7.5 Trapeze hangers 7.6 Duct hangers/straps 7.7 Shock mounts 7.8 Bulkhead supports Fasteners: 7.9 Hex bolts & nuts 7.10 Stud bolts 7.11 U-bolts 7.12 Eye bolts 7.13 Machine screws 7.14 Solid rivets 7.15 Blind (pop) rivets 7.16 Weld studs

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify different types of pipes based on material and usage. 1.2 Identify the steel pipe dimensions chart based on system requirements. 1.3 Describe the material characteristics of GI pipes. 1.4 Describe the material characteristics of MS pipes. 1.5 Identify different types of valves based on function and design.
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	1.6 Explain installation methods of valves following standard operating procedures. 1.7 Identify different types of pipe fittings based on shape and function. 1.8 Interpret fitting standards and specifications based on engineering drawings or manuals. 1.9 Identify incompatible or defective fittings. 1.10 Identify different types of pumps based on design and application. 1.11 Interpret pump specifications and operational symbols according to technical documents. 1.12 Explain pipe supporting and fastening methods. 1.13 Interpret standards and load capacities of supports and fasteners from technical guidelines.
4. Underpinning Skills	2.1 Performing identification of different types of pipes based on material and usage. 2.2 Demonstrating identification of steel pipe dimensions using system requirements. 2.3 Applying knowledge to describe material characteristics of GI and MS pipes. 2.4 Fixing the interpretation of pipe standards and markings in line with industry codes. 2.5 Maintaining identification of different types of valves based on function and design. 2.6 Servicing the explanation of valve functions according to technical specifications. 2.7 Performing installation methods of valves according to standard operating procedures. 2.8 Demonstrating identification of different types of pipe fittings based on shape and function. 2.9 Applying connection methods for fittings based on system requirements.
5. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of pipes 4.3 Different types of valves 4.4 Different types of pipe fittings 4.5 Different types of pumps

	4.6 Different types of hangers, supports, and fasteners
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified the different types of pipes 1.2 identified different types of valves 1.3 identified different types of fittings 1.4 identified different types of pumps 1.5 identified different types of hangers/support and fasteners
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM WELDING WORKS	Nominal Duration: 40 Hrs.	Unit Code: SICIP-SBD-SHP-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform welding works. It specifically includes the task of identifying types of welding and welding terminology, identifying different types of welding technique, carrying out arc welding and carrying out tack welding.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify types of welding and welding terminology	1.1. <u>Tools and equipment</u> required for welding operations are identified. 1.2. Different <u>types of welding</u> are identified and described according to application. 1.3. <u>Types of welding joints</u> are identified. 1.4. Common <u>welding terminology</u> are listed and defined clearly.
2. Identify different types of welding technique	2.1 Pipe welding <u>positional techniques</u> are identified and explained. 2.2 <u>Arc welding techniques</u> are identified and explained. 2.3 <u>Welding faults and tests</u> are identified and explained.
3. Carry out arc welding	3.1 <u>Personal Protective Equipment (PPE)</u> is collected and worn properly before welding. 3.2 Arc welding machine and tools are prepared and checked for safe operation. 3.3 Workpieces are positioned and secured according to welding instructions. 3.4 Arc welding is performed using correct electrode, technique, and parameters as per job requirement. 3.5 Weld joints are inspected for consistency, penetration, and quality.
4. Carry out tack welding	4.1 Workpieces are aligned and clamped in position as per job specifications. 4.2 Tack welds are applied at appropriate intervals to hold components securely before final welding. 4.3 Tack weld quality and positioning are checked to ensure correct alignment and fit-up. 4.4 Equipment is cleaned and stored after completion of tack welding tasks.

Range of Variables

Variable	Range (Includes but not limited to):
1. Tools and equipment	1.1 Arc welding machine 1.2 Electrode holder 1.3 Earth clamp 1.4 Chipping hammer 1.5 Steel wire brush
2. Types of welding	2.1 Arc 2.2 Gas
3. Types of welding joints	3.1 Butt joint 3.2 Tee joint 3.3 Lap joint 3.4 Corner joint 3.5 Edge joint
4. Welding terminology	4.1 Work piece 4.2 Electrode 4.3 Slag 4.4 Root 4.5 Root gap 4.6 Weld bead 4.7 Welding faults 4.8 Penetration 4.9 Electrode Code 4.10 Electrode holder 4.11 Edge preparation 4.12 Welding unit/transformer 4.13 Polarity 4.14 Alternating current 4.15 Direct current
5. Positional techniques	5.1 Flat position 5.2 Horizontal 5.3 Vertical 5.4 Overhead 5.5 Rolled /wrap around position (for different angles)
6. Arc welding techniques	6.1 Fillet 6.2 Tack 6.3 Groove
7. Welding faults and tests	7.1 Welding faults: 7.1.1 Porosity 7.1.2 Undercut 7.1.3 Slag inclusion 7.1.4 Blow hole 7.1.5 Lack in penetration 7.1.6 Weld crack 7.2 Tests: 7.2.1 Destructive

	7.2.2 Non-destructive
8. Personal Protective Equipment (PPE)	8.1 Apron 8.2 Hand gloves 8.3 Safety shoes 8.4 Welding helmet 8.5 Safety goggles 8.6 Skull protection

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify tools and equipment required for welding operations. 1.2 Identify and describe different types of welding according to their applications. 1.3 Identify types of welding joints. 1.4 Identify and explain pipe welding positional techniques. 1.5 Identify and explain arc welding techniques. 1.6 Identify and explain welding faults and tests. 1.7 Align and clamp workpieces. 1.8 Tack welds procedures. 1.9 Tack weld quality and positioning.
2. Underpinning Skills	2.1 Applying knowledge to identify and describe types of welding joints. 2.2 Fixing and listing common welding terminology with clear definitions. 2.3 Maintaining understanding of pipe welding positional techniques and arc welding methods. 2.4 Preparing and checking arc welding machine and tools for safe operation. 2.5 Securing and positioning workpieces according to welding instructions. 2.6 Carrying out arc welding using correct electrode, technique, and parameters as per job requirements. 2.7 Inspecting weld joints for consistency, penetration, and quality. 2.8 Applying tack welds at intervals to secure components before final welding. 2.9 Checking tack weld quality and positioning for correct alignment and fit-up.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace

	3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Tools and equipment 4.3 Welding machine

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified types of welding and welding terminology 1.2 identified different types of welding technique 1.3 carried out arc welding 1.4 carried out tack welding
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM PIPE FABRICATION WORKS	Nominal Duration: 120 Hrs.	Unit Code: SICIP-SBD-SHP-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform pipe fabrication works. It specifically includes the tasks of cutting, threading and beveling pipes, performing pipe bending with different angle, cleaning and testing piping and securing supports, hangers and guides.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Cut, thread and bevel pipes	1.1. <u>Personal Protective Equipment (PPE)</u> is collected and worn properly before pipe cut, thread and beveling. 1.2. Appropriate <u>tools and equipment</u> are identified and prepared. 1.3. Suitable <u>cutting techniques</u> are selected and applied according to pipe type and job conditions. 1.4. Pipes are cut as per dimensions and project specifications. 1.5. Threading, beveling, and grinding of pipe ends are performed in compliance with technical standards.
2. Perform pipe bending with different angle	2.1 <u>Pipe bending terms</u> are identified and defined. 2.2 Pipes are marked and measured according to the required bending angles. 2.3 Pipe bending machines are set up and operated following manufacturer instructions. 2.4 Pipes are bent to various specified angles without distortion, flattening, or wrinkling. 2.5 Completed bends are inspected and verified for accuracy, alignment.
3. Clean and test piping	3.1 <u>Cleaning and inspection</u> of the piping assembly are carried out to remove debris. 3.2 Appropriate <u>testing methods</u> are selected and applied as per specifications. 3.3 Test results are observed, recorded, and evaluated for compliance with standards. 3.4 Defects or leaks detected during testing are reported and rectified according to procedures. 3.5 Work area is cleaned and restored to a safe condition upon completion of piping fabrication and testing.
4. Secure supports, hangers and guides	4.1 <u>Pipe supporting, hanging, and fastening methods</u> are identified and explained. 4.2 Appropriate supports, hangers, and guides are selected according to pipe size.

	4.3 Installation positions are marked and verified before securing components. 4.4 Supports and hangers are fastened securely using proper tools and techniques as per job requirements. 4.5 Installed support systems are checked for stability, alignment, and compliance with safety standards.
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Range of Variables

Variable	Range (Includes but not limited to):
1. Personal Protective Equipment (PPE)	1.1 Apron 1.2 Hand gloves 1.3 Safety shoes 1.4 Safety goggles
2. Tools and equipment	2.1 Gas cutting sets 2.2 Grinding machine 2.3 Centre punch 2.4 Level bar 2.5 Ball pin hammer 2.6 Spirit level 2.7 Soft stone 2.8 Measuring tape 2.9 Files 2.10 Power brush 2.11 Steel brush 2.12 Adjustable wrench 2.13 Tip cleaner 2.14 Contour marker 2.15 Plumb bob 2.16 Spark lighter
3. Cutting techniques	3.1 Oxy-Acetylene Cutting 3.2 Plasma Cutting 3.3 Mechanical Sawing 3.4 Grinding Wheel Cutting 3.5 Cold Sawing 3.6 Band Saw Cutting
4. Pipe bending terms	4.1 Bend Radius 4.2 Centerline Radius (CLR) 4.3 Degree of Bend 4.4 Wall Thinning 4.5 Wrinkling 4.6 Bend Allowance 4.7 Neutral Axis 4.8 Compression Side 4.9 Tension Side
5. Cleaning and inspection	5.1 Chemical flushing 5.2 Water flushing

	5.3 Air flushing
6. Testing methods	6.1 Hydrostatic testing 6.2 Pneumatic testing 6.3 Non-destructive testing
7. Pipe supporting, hanging, and fastening methods	7.1 Pipe Clamp 7.2 Pipe Hanger 7.3 U-Bolt 7.4 Saddle Support 7.5 Pipe Shoe 7.6 Adjustable Support 7.7 Clevis Hanger 7.8 Rod Hanger 7.9 Wall Bracket 7.10 Strap Hanger 7.11 Pipe Rack

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify and explain the proper use of Personal Protective Equipment (PPE). 1.2 Identify and describe the appropriate tools and equipment needed for the task. 1.3 Identify cutting techniques based on the pipe type and job conditions. 1.4 Understand threading, beveling, and grinding of pipe ends. 1.5 Identify and define pipe bending terminology. 1.6 Understand pipe bending machines. 1.7 Understand testing methods based on the specifications. 1.8 Identify defects or leaks detected during testing according to standard procedures. 1.9 Identify and explain the methods for supporting, hanging, and fastening pipes.
2. Underpinning Skills	2.1 Demonstrating identification and preparation of appropriate tools and equipment for pipe cutting and threading. 2.2 Applying suitable cutting techniques based on pipe type and job conditions. 2.3 Fixing pipes to the specified dimensions and project specifications. 2.4 Maintaining technical standards while performing threading, beveling, and grinding of pipe ends. 2.5 Identifying and defining pipe bending terms and selecting the correct bending techniques. 2.6 Setting up and operating pipe bending machines following manufacturer instructions. 2.7 Bending pipes to the specified angles without distortion,

	flattening, or wrinkling. 2.8 Applying appropriate testing methods and evaluating test results for compliance with standards. 2.9 Selecting appropriate supports, hangers, and guides based on pipe size.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Tools and equipment 4.3 Required materials

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 cut, thread and bevel pipes 1.2 performed pipe bending with different angle 1.3 cleaned and tested piping 1.4 secured supports, hangers and guides
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM INSTALLATION OF PIPING SYSTEM	Nominal Duration: 70 Hrs.	Unit Code: SICIP-SBD-SHP-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform installation of piping system. It specifically includes the tasks of installing fresh water piping system, installing sea water piping system, installing fuel piping system and installing hydraulic piping system.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Install fresh water piping system	1.1. Drawings, tools, and equipment are identified and selected in accordance with the job requirements. 1.2. Pipes are measured and cut accurately as per the piping layout shown in the drawing. 1.3. Pipe ends are threaded or welded depending on the joining method as required. 1.4. <u>Appropriate fittings</u> are selected and fixed onto the fresh water pipes following installation standards. 1.5. Pickling and cleaning of piping spools are performed to remove contaminants. 1.6. Fresh water piping system is assembled and installed as per the specified job procedure. 1.7. Installed piping system is checked and confirmed to be free from contamination and ready for use.
2. Install sea water piping system	2.1 Drawings, tools, and equipment are identified and selected. 2.2 Pipes are cut within specified tolerances as per the approved drawings and job specifications. 2.3 Pipe ends are threaded or welded depending on the joining method as required. 2.4 Appropriate fittings are selected and fixed onto the sea water pipes following standard practices. 2.5 Sea water piping system is assembled and installed as per the technical and job requirements. 2.6 Completed piping assembly is inspected and confirmed to be free from contamination.
3. Install fuel piping system	3.1 Drawings, tools, and equipment are identified and selected according to job specifications. 3.2 Pipes are cut within specified tolerances as per the approved drawings and work instructions. 3.3 Pipe ends are threaded or welded using appropriate methods based on the joining requirement. 3.4 Suitable fittings are selected and fixed onto the fuel

	piping as per technical requirements. 3.5 Pressure testing of the piping spools is carried out to ensure system integrity and leak-proof installation. 3.6 Pickling and cleaning of piping spools are performed to remove contaminants. 3.7 Fuel piping system is assembled and installed in compliance with safety and performance standards. 3.8 Completed piping assembly is inspected and verified to be free from contamination before commissioning.
4. Install hydraulic piping system	4.1 <u>Components of the hydraulic piping system</u> are identified and understood as per system design. 4.2 Drawings, tools, equipment, and fittings are identified and selected according to the job requirement. 4.3 Appropriate fittings are selected and fixed onto the piping system using threading or welding as specified. 4.4 Pickling and cleaning of piping spools are carried out to remove contaminants. 4.5 Pressure testing of piping spools is performed to verify strength and leak-proof integrity. 4.6 Hydraulic piping system is installed and precisely as per job instructions and safety guidelines.

Range of Variables

Variable	Range (Includes but not limited to):
1. Appropriate fittings	1.1. Flange 1.2. Union 1.3. Elbow 1.4. Tee 1.5. Socket and bon plug 1.6. Nipple 1.7. Reducer
2. Components of hydraulic piping system	2.1 Reservoir/hydraulic cylinder 2.2 Strainers and filters 2.3 Accumulators 2.4 Pumps 2.5 Directional control valves

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identify and select drawings, tools, and equipment. 1.2 Identify fix appropriate fittings onto the fresh water pipes following installation standards. 1.3 Identify drawings, tools, and equipment for the sea water piping system. 1.4 Thread or weld pipe ends based on the required joining method.
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	1.5 Identify drawings, tools, and equipment according to job specifications for the fuel piping system. 1.6 Understand pressure testing of the piping spools to ensure system integrity and leak-proof installation. 1.7 Interpret the fuel piping system in compliance with safety and performance standards. 1.8 Identify and understand components of the hydraulic piping system as per system design. 1.9 Identify drawings, tools, equipment, and fittings according to job requirements.
2 Underpinning Skills	2.1 Performing the identification and selection of drawings, tools, and equipment. 2.2 Demonstrating accurate measurement and cutting of pipes as per the piping layout. 2.3 Applying threading or welding techniques to pipe ends based on the required joining method. 2.4 Fixing appropriate fittings onto the fresh water pipes following installation standards. 2.5 Fixing pickling and cleaning of piping spools to remove contaminants. 2.6 Assembling and installing the fresh water piping system according to specified job procedures. 2.7 Checking and confirming the installed piping system to be free from contamination and ready for use. 2.8 Performing the identification and selection of tools, drawings, and equipment. 2.9 Performing pressure testing on the piping spools to ensure leak-proof installation. 2.10 Performing pressure testing on hydraulic piping spools to verify strength and integrity. 2.11 Installing the hydraulic piping system according to job instructions and safety guidelines.
3 Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Appropriate fittings 4.3 Tools and equipment 4.4 Required materials

Assessment Evidence Guide

1 Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 installed fresh water piping system
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	1.2 installed sea water piping system 1.3 installed fuel piping system 1.4 installed hydraulic piping system
2 Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio review
3 Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	Ship Piping
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sq) OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

Sl. No.	Major Equipment and Training facilities	Required facilities
1.	Arc Welding Machine	05
2.	Grinding machine	10
3.	Pipe Bending Machine	01
4.	Fresh Water Piping System	01
5.	Sea Water Piping System	01
6.	Fuel Piping System	01
7.	Hydraulic Piping System	01
8.	Valves (gate valve, SDNR valve, check valve, ball valve)	05 sets each
9.	Minimum fittings (union, tee, socket, elbow, nipple)	05 sets each
10.	Pumps (centrifugal, gear)	05 sets each
11.	Tool box	05 sets

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Ship Piping the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.